Aadhaar OTP Request Flow

Aadhaar OTP authentication is used to verify the identity of an Aadhaar holder by sending a One-Time Password to the mobile number registered with Aadhaar. The authentication is successful only if the user enters the correct OTP within the allowed time

OTP Request Process:

Application initiated OTP request works as follows:

1. Application (an application on an assisted device or self-service kiosks, or applications on the Internet), wanting to use Aadhaar OTP as a factor within Aadhaar authentication, initiates the transaction flow.
2. Application captures Aadhaar number/Virtual ID (VID) along with any other data as needed.
3. Application, through AUA server, invokes the OTP Request API by forming digitally signed API Input XML (format explained later in this document).
4. UIDAI server processes the input, validates it, generates OTP, and sends it to resident's registered mobile/email.
5. UIDAI server then responds to OTP Request API with a response XML with success or indicating any error (see response XML details in later sections).
6. Resident receives the OTP from Aadhaar server on his/her email and/or mobile.
7. Application then requests resident to provide the OTP value so that application can send that via Aadhaar authentication API for authenticating the resident.
1. After receiving OTP, subsequent usage of OTP within Aadhaar authentication is explained in Aadhaar Authentication API specification 2.5. Notice that OTP expires with a UIDAI stipulated time. OTP message (SMS/Email) sent to resident provides time at which OTP was generated and when it is going to expire.

Prerequisites:

- AUA/Sub-AUA Code
- Valid UIDAI License Key
- UIDAI Public Certificate

OTP Generation Request:

API Protocol:

https://apisandbox.uidai.gov.in/uidotpserver/2.5

Note: Please obtain the Aadhaar OTP Authentication APIs from UIDAI Sandbox Portal

API Request Data Format:

Following is the XML data format for OTP API:

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>

<Otp uid="" ac="" sa="" ver="" txn="" ts="" lk="" type="">

<Opts ch=""/>

<Signature>Digital signature of AUA</Signature>

</Otp>

Note: Reach out to Auth support team for Digital signature

Attributes and Values:

uid=" Enter Sandbox UID"

ac=" public"

sa=" public"

ver="2.5"

txn=" Enter the transaction number"

ts=" Enter Time stamp in YYYY-MM DDThh:mm: ss format"

lk=" MG_g7jJVYUIW7cLYXY5yaqKD6D1TuhjTJTDPHcb0SudOhVpvpnsEw_A"

Digital signature of AUA: request XML should be digitally signed for message integrity and non-repudiation purposes

CI=20300811

Note: All the Authentication API, Certificates and configuration must be obtained from the UIDAI Sandbox Portal

Element Details Element:

Otp (mandatory)

- Root element of the input XML for OTP service.

Attributes:

- uid – (mandatory) –Aadhaar number (wherever Aadhaar number is used, in future, an encrypted Aadhaar number may also be used) or Virtual ID or UID Token or new mobile number of residents requesting OTP.
- ac – (mandatory) A unique code for the AUA which is assigned by UIDAI during AUA registration process. This is an alpha-numeric string having maximum length 10. (A Default value "public" is available only for testing.)
- sa – (mandatory) A unique "Sub-AUA" code. AUAs are expected to manage these codes within their system and ensure uniqueness within their system. This allows auditing and business intelligence to be provided at SA level. If AUA and SA are same agency, use value of "ac" for this attribute. This is an alpha-numeric string having maximum length 10.
- ver – (mandatory) version of the OTP API. Currently only valid value is "2.5".
- txn – (mandatory) AUA specific transaction identifier. AUA can choose to pass this as part of input. This is returned as part of response as is. This is very useful for linking transactions full round trip across systems. This is an alpha-numeric string of maximum length 50. Only supported characters are A-Z, a-z, 0-9, period, comma, hyphen, backward & forward slash, left & right parenthesis, and colon. No other characters are supported. It is highly recommended that AUAs use this attribute for correlating requests with responses for auditing and verification. Please refer to Authentication API 2.5 speciation to know about further usage of txn namespace when calling Authentication using OTP.
- ts – (mandatory) Timestamp at the time of request. This is in format "YYYY-MM DDThh:mm: ss" (derived from ISO 8601). Time zone should not be specified and is automatically defaulted to IST (UTC +5.30). If the UDIAI server receives the request after 20 min from this value, it is rejected.
- lk – (mandatory) A valid "License Key" assigned to the AUA. Administration portal of UIDAI will provide a mechanism for AUA administrator to generate license keys. This is an alpha-numeric string of length up to 64 characters.

API Response Data Format Response XML is as follows:

<OtpRes ret=” y” code=”” txn=” OTP-Req-001” err=”” ts=”” info=””/>

2.5.1 Element Details

Element: OtpRes

Attributes:

- ret – Result of OTP generation request. “y” if successful, “n” if failure in which case “err” attribute will provide the reason of failure.
- code – Unique alphanumeric “OTP response” code having maximum length 40.
- txn – AUA specific transaction identifier. This is the same value that is sent within the request.
- ts – Timestamp when the response is generated. This is of type XSD date Time.
- err – Failure error code. If OTP request fails, this attribute provides any of the following codes: o “110” – Aad